Shoaib Shaikh

Senior Software Engineer - +1 (647) 469-6373 - New York Metropolitan Area

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Summary

Senior software engineer who leads cross-team technical strategy for high-scale advertising systems, turning ambiguous problems into clear, shippable direction. 7+ years across the full ad stack (creative, targeting, forecasting, pacing, delivery, and attribution), driving multi-million-dollar impact on revenue, cost savings, and compliance.

Experience

Amazon – Amazon Ads July 2025 – Present

Software Development Engineer II (Tech Lead) *New York, NY, USA*

- Serving as technical lead for Kindle ads engineering, owning the technical roadmap end-to-end and directing migration and deprecation of the legacy ad server with zero service disruption to advertisers.
- Delivered a $2M annualized revenue increase (20%+ delivery improvement) by architecting an emergency migration off a deprecated targeting service under a hard deadline, discovering and shipping a billable ad-matching optimization for advertisers across the platform.
- Led technical design and presented Kindle's integration into Amazon's cross-platform ad-pacing infrastructure at a multi-team design review, resolving partner-team concerns, and ran a cross-team research spike across 6 teams to assess a new Programmatic Guaranteed deal product on Kindle; findings informed another team's org-wide strategy document.
- Designed and delivered end-to-end customer-based personalization signal spanning new data pipelines, feature onboarding, and a new runtime fetcher across 5 services - consulting with multiple engineering teams and applied scientists - unblocking a company-wide personalization initiative for ad relevance.
- Oversaw a Kindle-wide Java runtime migration, scoping which devices required it, authoring the fleet-wide validation standard, unblocking a peer engineer's cutover change, and coordinating JDK upgrades across four dependent services. Also migrated three production services off a hard-deprecating legacy backend, re-architecting mid-flight after a shared-dependency issue, with zero disruption and a 30% latency improvement.
- Fixed two high-impact delivery and reliability failures: diagnosed a silent ad-delivery failure blocking a next-generation e-reader device's launch, with no logs or errors pointing to a cause, through investigation across app, device, and configuration layers; and root-caused a SEV-2 incident spanning two regions to a misconfigured internal throttle rule, leading cross-team resolution, and shipped a durable fix to prevent recurrence.
- Reduced infrastructure costs by identifying hardware and infrastructure savings with fleet owners and leadership, and by leading a migration from RDS to DynamoDB prompted by a forced MySQL 5.7-to-8 upgrade that would have broken backend services if left unaddressed, cutting infrastructure costs 60% .
- Led delivery of European Accessibility Act (EAA) compliance across all Kindle ad placements, including a full legacy template refactor and gap-analysis review with accessibility, product, and legal teams and partners.
- Served as a trusted cross-org point of contact for extending forecasting support across 6 partner teams spanning Alexa, Fire Tablet, and Prime Video, leading requirements and implementation for new ad products, and provided ad campaign performance guidance to advertising sales leadership, investigating under-performing campaigns and publishing troubleshooting guides used directly by account teams across the organization.
- Adopted agentic AI tooling into daily engineering practice – driving automated dead-code detection and cleanup across services and accelerating delivery on multiple migrations and onboardings across teams.
- Provide technical and career mentorship to 10+ engineers and interns, supporting promotions, return offers, and improved performance ratings for mentees.

Humber Polytechnic January 2023 – August 2025

Adjunct Professor *Toronto, ON, Canada*

- Instructed core CS courses (Unix System Administration, Web Development, JavaScript, and Python) to 1000+ students, and created learning materials using practical, hands-on projects and real-world examples for students.

Amazon – Amazon Ads October 2021 – July 2025
Software Development Engineer II *Toronto, ON, Canada*

- Built a general-purpose asynchronous, multi-threaded enrollment pipeline, increasing device processing throughput 10x; applied it to safely unenroll 12M devices from a legacy ad program, with headroom for larger-scale future migrations across the fleet for devices
- Eliminated critical hot-partition bottlenecks by migrating to multi-shard AWS Kinesis, stabilizing high-volume traffic.
- Refactored privacy-sensitive decade-old legacy systems under tight deadlines to achieve Digital Markets Act (DMA) compliance with zero downtime
- Led a database right-sizing initiative via blue-green deployments, cutting annual cloud costs 80% ($1.2M)
- Led launch of ad forecasting across Alexa, Fire Tablet, and Prime Video, enabling accurate impression prediction
- Onboarded two new e-reader device lines to ads platform on schedule despite significant technical obstacles, reverse-engineering deprecated ad-package generation tooling and resolving on-device memory issues – generating $2M+ in annual revenue for the platform
- De-risked production targeting-service migration by designing multi-step shadow-call validation strategy, and authored initial implementation that peer engineer extended to complete migration

Amazon – Amazon Ads February 2020 – October 2021
Software Development Engineer I *Toronto, ON, Canada*

- Architected and delivered the compatibility layer that decoupled Kindle's decade-old legacy ad-server from device software, enabling faster and safer ad updates
- Expanded advertising services from US-only to global marketplaces, enabling international scale
- Implemented React web components shown to 100K+ daily active users

The Citizen Lab – University of Toronto January 2020 – April 2020
Research Assistant *Toronto, ON, Canada*

- Developed ETL pipelines to collect, clean, and analyze large-scale online datasets for academic research

Amazon – Fulfillment by Amazon (FBA) May 2019 – August 2019
Software Development Engineer Intern *Toronto, ON, Canada*

- Reduced API latency from 500ms to 10ms by eliminating AWS Lambda cold starts via S3-backed caching
- Improved storage performance by regionalizing S3 buckets and optimizing integration tests
- Built internal full-stack debugging tool to replace manual log inspection, slashing developer troubleshooting time

IBM May 2018 – May 2019
Software Developer Intern *Markham, ON, Canada*

- Reduced automated end-to-end test runtime by 90% via a fail-fast testing strategy and improved CI/CD visibility with automated Slack/email reporting tools

Education

Georgia Institute of Technology Atlanta, GA, USA
M.S. in Computer Science (Specialization in Computing Systems) – Online *September 2026 – Present*

University of Toronto Toronto, ON, Canada
Honours B.Sc. in Computer Science (Specialist Focus in Artificial Intelligence), Minor in Economics *September 2019 – April 2022*

Technical Skills

Languages: Java, Python, JavaScript/TypeScript, C/C++, HTML/CSS, Kotlin, PHP, Ruby, SQL
Cloud: AWS (Lambda, DynamoDB, S3, SQS, SNS, RDS, EC2, ECS, Kinesis, Bedrock, IAM, VPC), Azure
Frameworks/Runtimes: Node.js, Express, React, Next.js, Spring Boot, Vue, Angular, Shadcn, Tailwind CSS
AI/ML & Data: Generative AI (LLMs, Agentic AI), OpenAI/Claude/Gemini/Grok APIs, PyTorch, TensorFlow, pandas, NumPy, scikit-learn
Tools & Platforms: Docker, Git, Jenkins, CI/CD, Supabase, Firebase, Stripe, Cursor
Concepts: Digital Advertising, Distributed Systems, Microservices, API Design, Operational Excellence, Technical Debt Reduction